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Exfoliation of layered double hydroxides (LDHs): a new route to mineralize atmospheric CO₂†

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Layered Double Hydroxide (LDH) of Mg–Al–Acetate (Mg/Al = 3) was prepared by an anion exchange method starting from a nitrate precursor. The high hydration energy of acetate was exploited to exfoliate LDH into nanosheets having a thickness of 2–3 nm. A colloidal suspension of LDH layers was freeze dried to retain the layers in the exfoliated state. Thermal decomposition of exfoliated LDH at 400 °C resulted in the mixed metal oxide having a surface area of 275 m² g^{−1} which is higher than pristine LDH oxide (100 m² g^{−1}). This enhanced surface area of exfoliated LDH is responsible for a large amount of CO₂ sorption (1.4 mmol g^{−1}) at room temperature as compared to pristine LDH (0.63 mmol g^{−1}). This enhanced CO₂ uptake is equal to some of the porous solids at similar conditions. The combination of aqueous exfoliation and the freeze drying technique has resulted in the successful synthesis of a novel CO₂ adsorption sorbent based on LDHs.

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1. Introduction

Carbon dioxide is a major green house gas and has a direct impact on global warming and climate change. At present, the concentration of CO₂ in the atmosphere is around 400 ppm and is rising exponentially over the decades.¹ Flue gases from coal based power plants and exhaust from automobiles are the major contributors to the rise in the CO₂ level.² This rise in atmospheric CO₂ level is a growing concern for the survival of the ecosystem. Therefore bringing about a significant reduction of CO₂ levels in the atmosphere is a major scientific challenge.

Two approaches are used to reduce the concentration of CO₂ level and thereby mitigate climate change and global warming: (i) carbon dioxide capture and storage (CCS)³ (ii) carbon dioxide capture and utilization (CCU).⁴ In the latter technique the captured CO₂ is used as a source of carbon in the production of fine chemicals.⁵ Over the years various solvent based and solid based sorbents have gained attention for capturing CO₂ at different temperatures, starting from the ambient conditions to elevated temperatures (800 °C).⁶ At present, amine scrubbing is one of the major industrial technique to capture CO₂ by using liquid amines such as 2-aminoethanol (MEA).⁷ The solvent based sorbents have numerous limitations such as (a) large amount of energy required for regeneration, (b) corrosive nature, (c) high cost of operation, and (d) limited temperature range of operation. In addition to this the degradation products

of amines such as nitrosamine and nitramines are known to be carcinogenic in nature.⁸ On account of these difficulties there is a pressing need for the replacement of solvent based sorbents by solid sorbents.

Numerous solid based sorbents such as metal organic frameworks (MOF),⁹ zeolites,¹⁰ activated carbons,¹¹ organic polymers,¹² alkali and alkaline based metal oxides,¹³ inorganic membranes,¹⁴ LDH based oxides¹⁵ and, various other solids¹⁶ have been proposed as candidate materials due to their potential for CO₂ capture applications. Among these, oxides derived from layered double hydroxides (LDHs) are potential sorbents for CO₂ capture in the temperature range of 200–400 °C since (i) they have good adsorption/desorption kinetics, (ii) large sorption capacity for CO₂, (iii) ideal systems to employ at flue gas conditions and, (iv) are environmentally benign.¹⁷

CO₂ sorption behaviour of LDHs/LDH based oxides have been studied by varying metal cations in the layer,¹⁸ anions in the interlayer¹⁹ and by the inclusion of alkali metal ions as promoters.²⁰ LDHs have also been dispersed on various supports like graphene oxide,²¹ carbon nanotubes,²² carbon nanofibres²³ and, zeolites²⁴ to study the effect of supports on CO₂ uptake. In addition to this the effect of synthetic conditions of the precursor LDHs²⁵ and, the role of water vapour on CO₂ sorption have been studied.²⁶ Despite all these efforts, the uptake of CO₂ by LDH based oxides is not significant enough due to (a) the low surface area of the oxides, (b) poor thermal stability and, (c) agglomeration of particles during thermal cycling.

Most of the studies involving LDH based oxides were aimed at capturing CO₂ at high temperature and regeneration of the oxide for reuse. Alternatively, LDHs can be developed as sinks for mineralizing CO₂ at ambient temperature as these materials

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have a high theoretical CO₂ uptake capacity. But no attempts are made in the literature in this direction.

In this paper we increase the surface area of the LDH oxides by exfoliating the precursor LDH layers. The resultant oxide residue exhibited better state of art capacity of LDH based oxides to mineralize CO₂ at the ambient temperature.

2. Experimental section

Synthesis

All reagents were purchased from Aldrich and used without further purification. Decarbonated type II water was used (specific resistance 15 MΩ cm, Millipore Elix-3 water purification system) throughout the synthesis. Decarbonation of water was done by boiling water for 30 min. and then passing N₂ gas to remove dissolved CO₂. Mg–Al–NO₃ LDH was synthesized by employing ammonia precipitation method²⁷ in which the mixed metal nitrate solution (Mg/Al = 3) was added drop wise to a reaction vessel containing ammonia solution (100 mL, 1 M). Mg–Al–CH₃COO LDH was prepared by anion exchange reaction starting with Mg–Al–NO₃ precursor. A 10 times excess of the acetate in the form of sodium acetate was used to affect the anion exchange. Exfoliation of the Mg–Al–CH₃COO LDH was achieved by dispersing 250 mg of sample in 250 mL of water. The resultant suspension was stirred for 2 days on a magnetic stirrer. The resultant colloidal suspension was freeze dried. A mixed metal oxide residue was obtained by decomposing the freeze dried LDH (400 °C, 4 h, flowing N₂).

Characterization

All samples were characterized by powder X-ray diffraction (PXRD) using a PANalytical X'pert pro X-ray diffractometer (θ – 2θ scan, step size 0.017° 2θ , Cu K α radiation, λ = 1.541 Å, Bragg–Brentano geometry) equipped with a X'celerator scientific RTMS. Infrared spectra (IR) of the samples were recorded using a Perkin Elmer 100 FT-IR spectrometer in the UATR (universal attenuated total reflectance) mode. Elemental analysis (EA) was carried out on a Vario Elementar EL III apparatus. Thermal behaviour of the LDHs was studied using Mettler Toledo TGA/SDTA Model 851^e driven by star^c software (heating rate 5 °C min^{−1}, flowing N₂). The anion content was determined by ion chromatography (IC) using a Metrohm 861 Advanced Compact ion-chromatograph fitted with a Metrosep A SUPP5 250 anion column and conductivity detector. Topographical images of the exfoliated LDHs were recorded using a MFP3D™ Atomic Force Microscope (Asylum Research, Santa Barbara, California) equipped with silicon cantilevers (silicon tip, type NSC15/A1BS, μmash, Tallin, Estonia). A few drops of the LDH-suspension were dropped on to a silicon wafer previously cleaned with CO₂ snow-jet and spin coated at 2000 rpm. The scan rate was 1 Hz. CO₂ and N₂ sorption measurements were carried out on a Quantachrome Nova surface analyzer. All the samples were degassed in N₂ atmosphere for 24 h before the measurements were done.

3. Results

Composition of the Mg–Al–CH₃COO LDH was arrived at by independently estimating the metal contents (atomic absorption spectroscopy), acetate content (ion chromatography) and water content (thermogravimetric analysis). The measurements were mutually consistent and yielded an approximate formula [Mg_{0.73}Al_{0.27}(OH)₂] (CH₃COO)_{0.25}(CO₃^{2−})_{0.01}·0.8H₂O for the LDH.

PXRD pattern and IR spectra of starting Mg–Al–NO₃ LDH is given in ESI 1 and 2† respectively. The PXRD pattern of the Mg–Al–CH₃COO LDH (Fig. 1a) and shows a basal spacing of 12.2 Å (7.2° 2θ), which corresponds to the (001) reflection of the acetate-intercalated LDH, wherein the intercalated acetate is hydrated.²⁸ The higher order basal reflections, (002), and (003) are seen at 6.1 Å (14.2° 2θ) and 4.2 Å (21.2° 2θ) respectively. The turbostratic disorder as shown by the 'saw tooth' profile of reflections appearing in the mid 2θ region (35–50° 2θ) is due to the disorderly insertion of water molecules in the interlayer galleries of the acetate-LDH. The presence of acetate in the interlayer is indicated by IR spectra (Fig. 2a). The absorptions at 1552 and 1405 cm^{−1} are due to the antisymmetric and symmetric stretching vibrations of the COO[−] group of the acetate ion.²⁹ Acetate ion has a large hydration enthalpy and is responsible for swelling of acetate-intercalated LDHs when immersed in water.³⁰ The extreme manifestation of swelling results in exfoliation/delamination of metal hydroxide layers. Mg–Al–CH₃COO LDH when soaked in water exfoliated resulting in a stable colloidal suspension of metal hydroxide layers. The extent of exfoliation of the metal hydroxide layers was quantified by AFM measurements. Exfoliation resulted in nanosheets having thickness in the range of 2–4 nm and lateral dimensions in the range 100–200 nm (Fig. 3). Exfoliation of acetate intercalated Mg–Al LDH was further confirmed by doing Tyndall light scattering experiment on colloidal suspension (ESI 3†). Exfoliation leads to the high surface area material by prising apart successive layers from one another. When the excess water is removed, the layers restack to yield the original LDH, with a

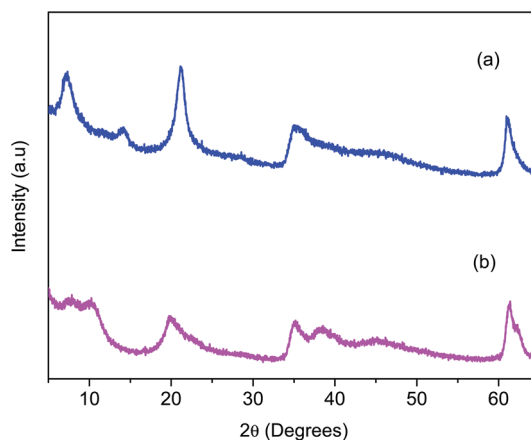


Fig. 1 PXRD pattern of Mg–Al–CH₃COO LDH (a) pristine LDH (b) freeze dried LDH.

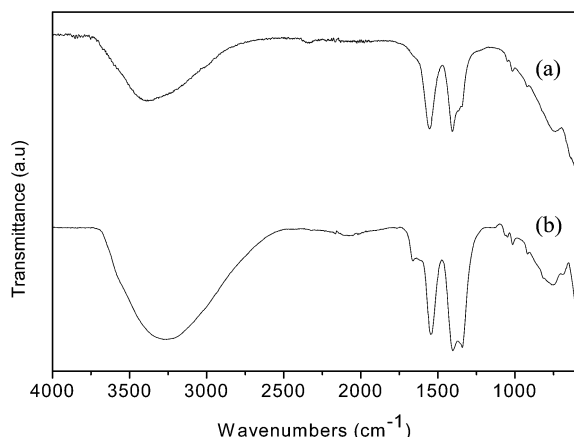


Fig. 2 IR spectra of Mg–Al–CH₃COO LDH (a) pristine LDH (b) freeze dried LDH.

concomitant reduction in the surface area. To prevent restacking and retain layers in exfoliated state, the colloidal suspension of the Mg–Al–CH₃COO LDH was freeze dried. PXRD pattern of freeze dried sample (Fig. 1b) shows first basal reflection at 11.2 Å (7.9° 2θ). In addition to this, a new peak at 8.6 Å (10.3° 2θ) is observed and its higher order reflection is seen at 4.4 Å (19.9° 2θ). These two reflections are due to (001) and (002) reflections of the dehydrated phase of acetate intercalated LDH. To check the continued presence of LDH phase after exfoliation and freeze drying IR spectra of the same has recorded and is given in Fig. 2b. Due to exfoliation, the bonding between the successive metal hydroxide layers along the stacking direction is lost. During freeze drying, some of the metal hydroxides layers have restacked but the extent of restacking is minimal as is evident in the reduced intensity of the basal reflections of the freeze-dried LDH. To see the effect of freeze drying on the suppression of restacking, we have exfoliated acetate LDH as before and restacked without freeze drying. From the PXRD pattern of restacked LDH without freeze drying is (ESI 4†) clearly shows the significance of freeze drying on the restacking of LDH layers.

LDHs decomposed at 400 °C under nitrogen atmosphere for 4 h are known to give mixed metal oxides having maximum CO₂ sorption capacity of 0.49 mmol g^{−1}.³¹ In an attempt to improve on this, the freeze dried LDH was decomposed (400 °C, 4 h, flowing N₂) and the CO₂ uptake capacity of the resultant oxide residue (hereinafter referred to as the test sample) was compared with that obtained by the decomposition of the pristine LDH (control sample). The PXRD patterns of the test and control samples (Fig. 4) show that in both cases, the oxide products are amorphous and exhibit broad humps around 43° 2θ, where the (200) reflection of MgO is expected. The full width at half maxima (FWHM) of the (200) reflection of MgO of the test sample, 3.1° 2θ is greater than that of the control at 2.2° 2θ. This indicates that smaller crystallites are present in the test sample. The exfoliation and freeze drying of LDH has prevented the particle agglomeration in the control to a large extent, resulting in a uniform distribution of smaller oxide crystallites.

To assess the surface properties such as pore volume, pore size and surface area of the oxide residues, N₂ adsorption/desorption experiment was carried out at 77 K. Prior to measurement both the oxide samples were degassed at 150 °C to remove adsorbed species. N₂ adsorption isotherm of both the oxides (Fig. 5) shows a Type IV isotherm. Pore condensation in both the isotherms indicates that the test and control oxides are both mesoporous in nature. Both the adsorption isotherms exhibit H2 type hysteresis indicating a disordered distribution of pore size and pore volume.³² Surface area of the oxides was calculated by multipoint BET method using the desorption branch of the isotherm. Pore volume and pore size were calculated by the HK method and BJH method respectively. The surface area of the freeze dried test oxide is higher at 275 m² g^{−1} compared to that of the control oxide which is 100 m² g^{−1} (Table 1). Similarly there is little increase in pore volume for freeze dried sample. The increased surface area and pore volume of the test sample could be accounted for by the exfoliation of precursor hydroxide layers.

Carbon dioxide uptake/adsorption behaviour of LDH oxide residues was studied by the gas adsorption technique by exposing the samples to CO₂ (25 °C, 1 bar). To avoid the adsorption of moisture, CO₂ and other gases present in the atmosphere, freshly decomposed LDH residues were used for the analysis. Control oxide residue shows 0.63 mmol g^{−1} of CO₂ adsorption. This value is in line with earlier reports for the LDH oxides.¹⁵ On the other hand, the test oxide residue obtained from the exfoliated LDH shows a CO₂ uptake of 1.4 mmol g^{−1} of CO₂ (Fig. 6). This large increase in CO₂ uptake shown by the test sample obtained from the exfoliated LDH is equal to the some of the porous solids at similar conditions.³³ This increase in CO₂ uptake by LDH based mixed metal oxides can be attributed to the high surface area of the oxide residue resulted from the exfoliation and freeze drying of the LDH layers.

4. Discussion

LDHs are also known as anionic clays and their structure consists of positively charged layers having the layer composition [M_{1−x}²⁺ M_x³⁺ (OH)₂]^{x+}. Charge compensating anions along with water molecules are get incorporated in to the interlayer galleries.³⁴ LDHs have a general formula [M_{1−x}M_x'(OH)₂](A^{n−})_{x/n}·mH₂O (M = Mg, Ca, Zn, Ni, Co, Cu; M' = Al, Fe, Cr, Ga and In; 0.15 ≤ x ≤ 0.33 and 0.5 ≤ m ≤ 1.0).³⁵ A number of naturally occurring LDHs are reported in the literature such as hydrotalcite, manasseite, meixnerite, pyroaurite, sjogrenite, coalingite, takovite to name a few. They all have intercalated carbonate as the counter ion.³⁵ The most common mineral among these is 'hydrotalcite' having the composition Mg₆Al₂(OH)₁₆CO₃·4H₂O (x = 0.25). There are two reasons for the ubiquitous presence of carbonate ions in all these minerals: (a) the molecular symmetry of the CO₃^{2−} (D_{3h}) compatible with the local symmetry of the trigonal prismatic interlayer site³⁶ and, (b) the hydroxides have a high affinity for CO₂ drawn from the ambient and mineralized as the CO₃^{2−} ion.

Among the various physicochemical properties, thermal behaviour of LDHs is very interesting. Thermal decomposition

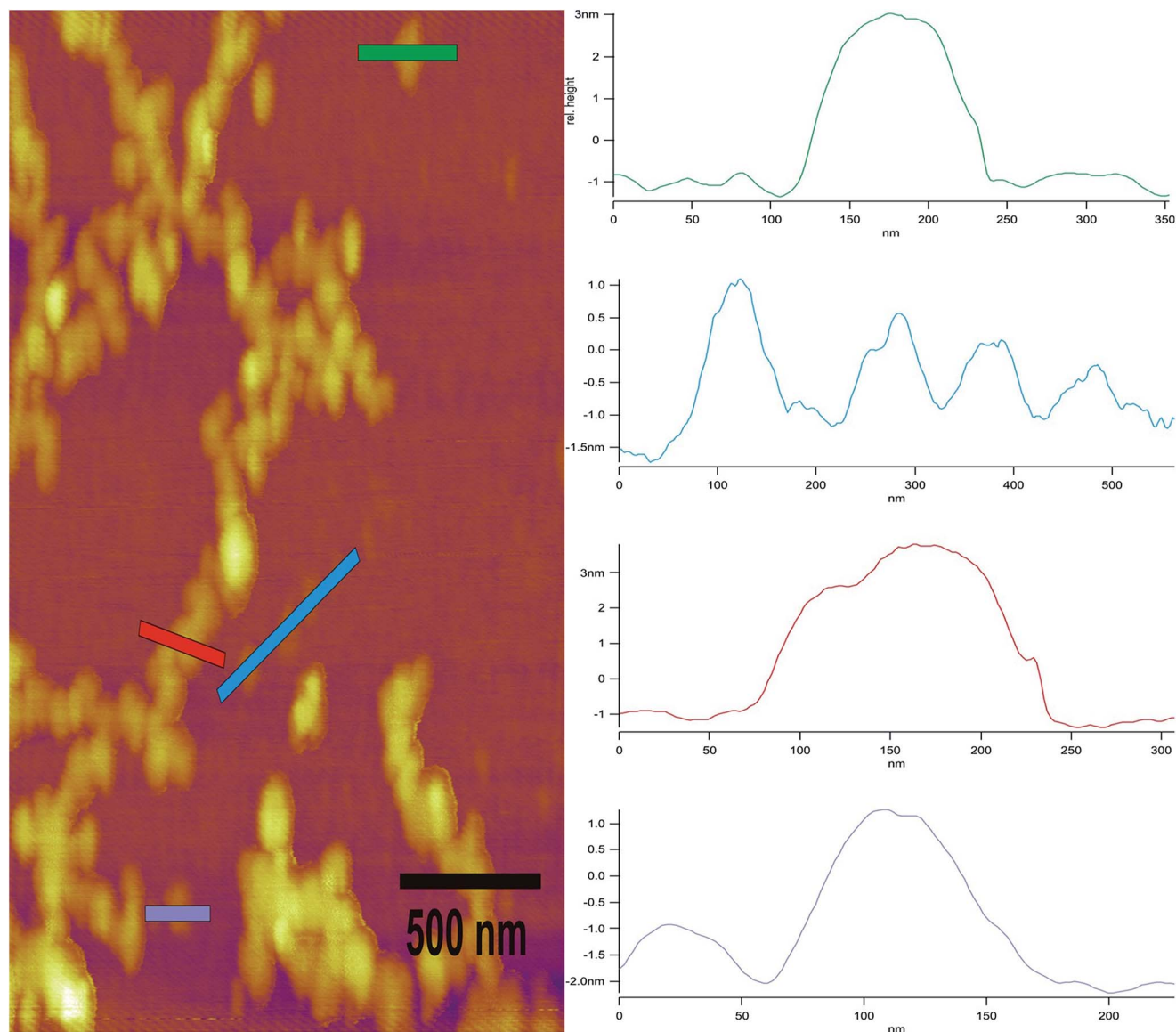


Fig. 3 (a) AFM topographical image of representative platelets of the Mg–Al–CH₃COO LDH. (b) The height profiles corresponding to the platelets in (a).

of LDHs around 400–450 °C yields mixed metal oxides. These oxides upon exposure to ambient or soaking in the aqueous solution containing anions reconstruct the parent LDH, a phenomenon called ‘memory effect’.³⁷ The mixed metal oxides obtained from thermal decomposition of LDHs are themselves useful base catalysts for various reactions.³⁸ Another application is sorption of CO₂ gas, arising due to the interaction of the basic metal oxide residues with acidic CO₂ to give metal carbonates,³⁹ which are in turn slowly converted into LDHs. The extent/amount of CO₂ sorption depends on the availability of basic sites on the oxide surface. In other words, amount and availability of the divalent metal oxide determines the extent of CO₂ sorption. The basic character of the mixed metal oxides obtained from thermal decomposition of LDHs can be tuned by varying the layer composition. For instance, the LDH with composition [Mg_{0.67}Al_{0.33}(OH)₂] (CO₃²⁻)_{0.165}·0.5 H₂O has ~7.6

mmol of Mg²⁺ for each gram of LDH and [Mg_{0.85}Al_{0.15}(OH)₂] (CO₃²⁻)_{0.075}·0.77 H₂O composition has ~11 mmol g⁻¹ of Mg²⁺. When all the Mg²⁺ present in the oxide residue is made available for sorption, these materials could be developed as potential sinks for capturing the large amount of CO₂. But contrary to the impressive theoretical CO₂ uptake capacity, the LDH residues sorbs only 0.5 to 0.6 mmol g⁻¹, leaving an immense head room for improvement. The low observed uptake capacity is primarily due to the lack of availability of basic sites.

To increase the number and availability of basic sites, one needs to break down the shared octahedron of hydroxyl ions within the layer and as well as break down the bonding between the successive layers to get nanoparticles of LDHs/LDH based oxides. To realize this one needs to design and develop new methodologies, which include new synthetic methods,⁴⁰ modified interlayers^{19a} and/or novel methods of exfoliation of the

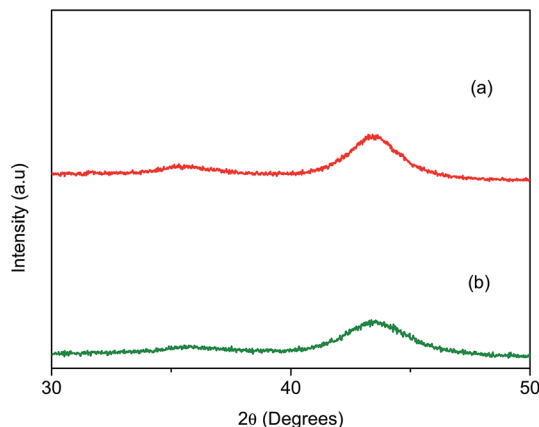


Fig. 4 PXRD pattern of oxide residue obtained from (a) pristine LDH and (b) freeze dried LDH.

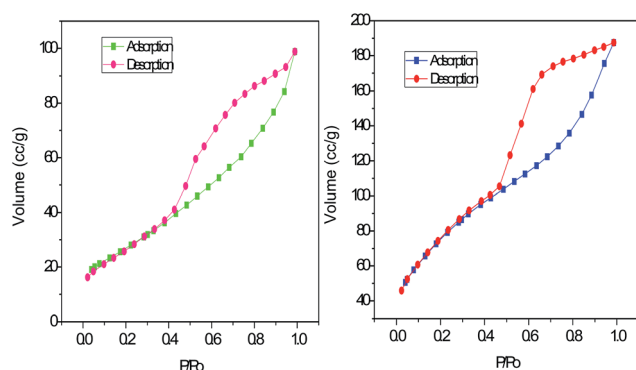


Fig. 5 N_2 adsorption isotherms of oxide residue obtained from (a) pristine LDH and (b) freeze dried LDH.

Table 1 Surface properties of exfoliated and pristine LDH oxide residues

	Surface area ($m^2 g^{-1}$)	Pore volume ($cm^3 g^{-1}$)	Pore diameter ($\text{\AA}$)
Pristine LDH	100.05	0.15	40.8
Exfoliated LDH	275.7	0.23	40.0

metal hydroxide layers.⁴¹ LDHs are known to undergo exfoliation/delamination in which metal hydroxide layers are pried apart to generate nanosheets.

5. Conclusions

To summarize, LDHs have a huge potential to capture and mineralize large amount of CO_2 for the following reasons:

(i) high affinity for carbonate ions, (ii) generation of basic oxides upon thermal treatment, (iii) basicity of the layers can be tuned by varying the layer composition, (iv) memory effect, (v) facile aqueous exfoliation to generate nanoparticles, (vi) the beneficial effect of adsorbed water/water vapour on CO_2 sorption, and (vii) they are environmentally benign and economical.

This work utilizes all of the above characteristics to obtain the highest reported sorption capacity for the LDHs. Firstly, the interlayer of the Mg–Al LDH was modified by the inclusion of a hydrophilic acetate anion by anion exchange. Secondly, the high hydration energy of acetate was successfully utilized to exfoliate the LDH into a few nanometre thick tactoids. The exfoliated tactoids were stabilized by freeze drying which suppressed restacking of the tactoids into thicker crystallites. Exfoliation coupled with freeze drying was successful in destroying the periodicity along the stacking direction to a larger extent. The effect of this is clearly seen in the distribution of particles and surface area of the oxide residue obtained after thermal decomposition. CO_2 sorption data further strengthens the role of exfoliation and freeze drying on the LDHs. In the present study, the exfoliation and freeze drying has not exposed all the Mg^{2+} present in the mixed metal oxide to CO_2 to the theoretically possible maximum. Nevertheless, it opens up the novel route to develop these materials as potential sinks for mineralizing CO_2 .

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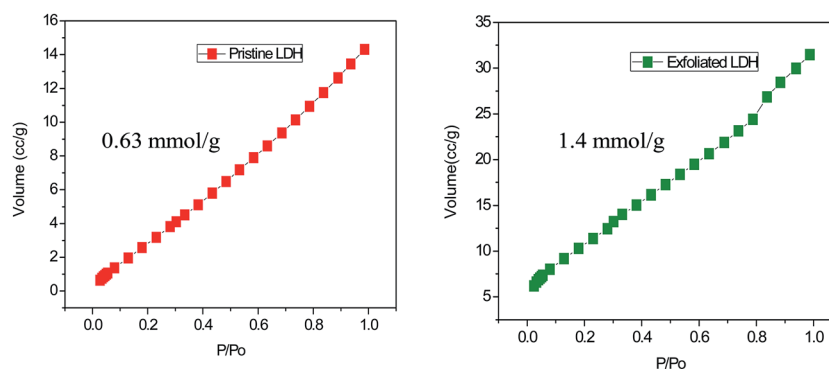


Fig. 6 CO_2 adsorption isotherms of oxide residue obtained from (a) pristine LDH and (b) freeze dried LDH.

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